

Experiment 6

6.1.1 Incremented Date

Algorithm :

Step 1: Start

Step 2: Input date, month, year

Step 3: valid_date = True.

Step 4: If (year $\leq$ 0 or month $>$ 12 or month $<$ 1)
then valid_date = False

Step 5: If month is 1,3,5,7,8,10,12
then max_date = 31
else if month is 4,6,9,11
then max_date = 30
else

if year $\%$ 4 == 0
max_date = 29
else
max_date = 28

Step 6: If (date $>$ max_date or date $<$ 1)
then valid_date = False

Step 7: If valid_date = True
if date $<$ max_date
date = date + 1
else
date = 1
if month == 12
month = 1
year = year + 1
else
month = month + 1

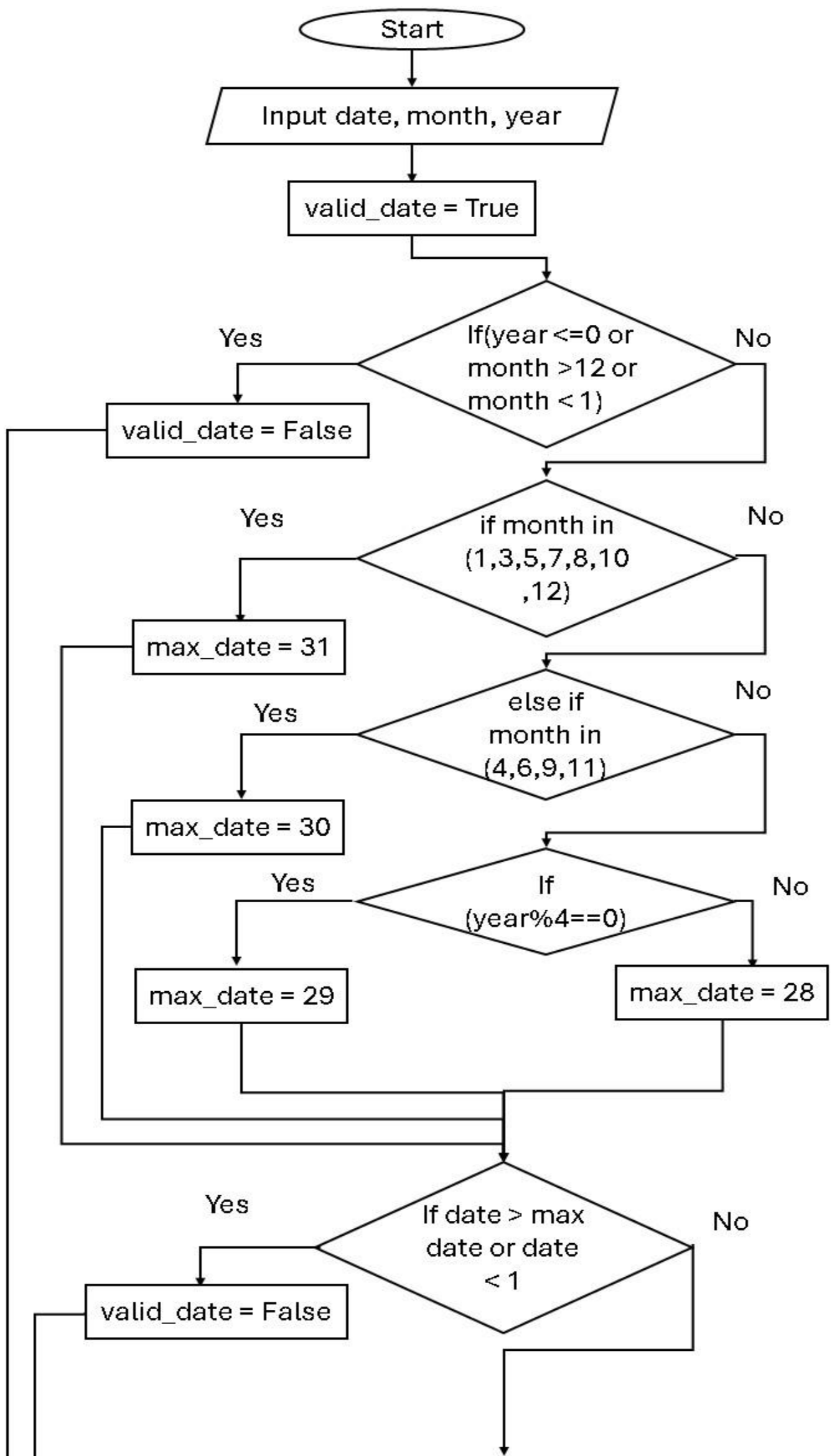
Step 8 :Print next date (date-month-year)

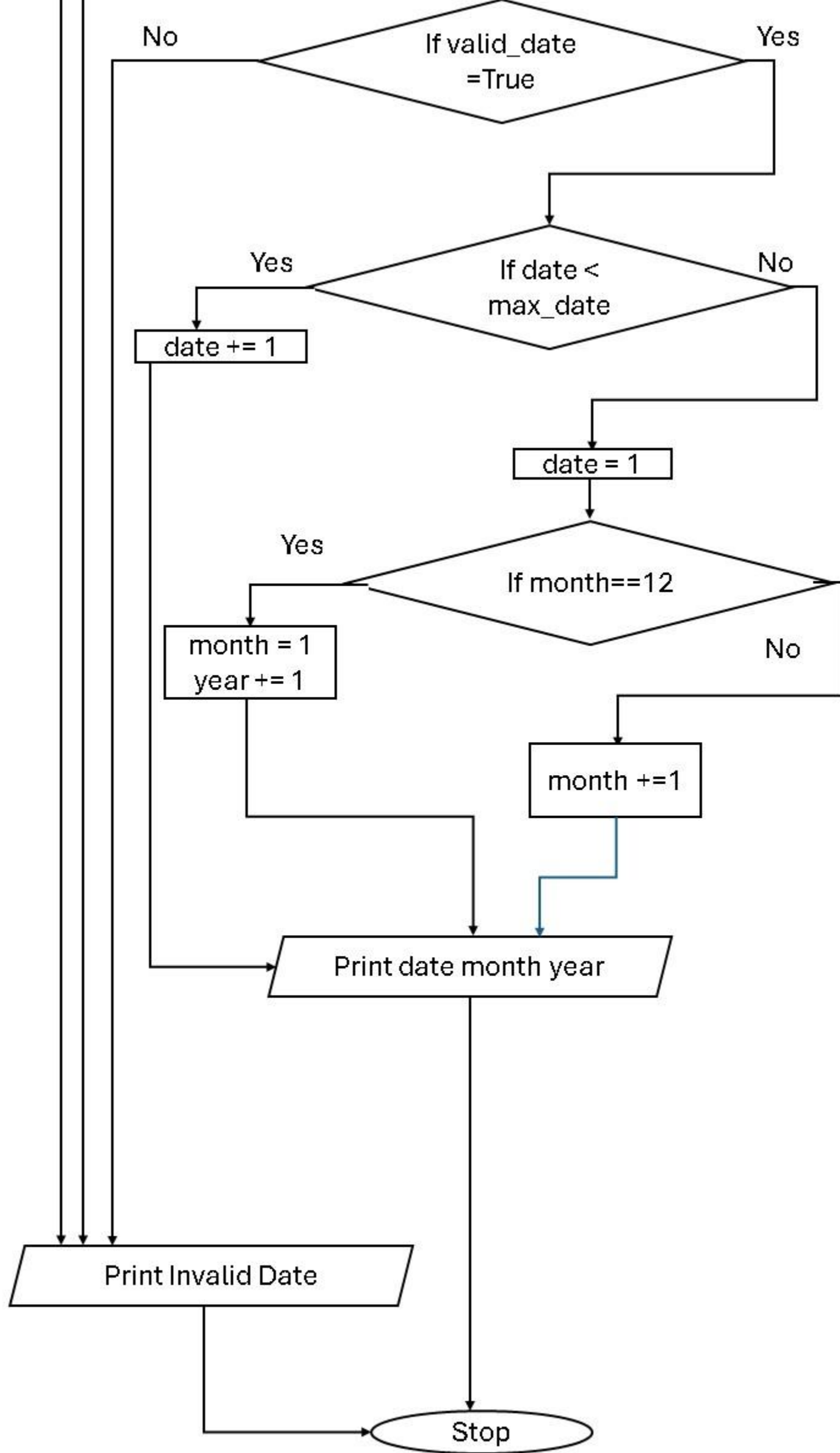
Step 9: Else

Print "Invalid Date".

Step 10: Stop

Flowchart :





Code :

```
date = int(input())
month = int(input())
year = int(input())

valid_date = True

if (year <= 0 or month > 12 or month < 1):
    valid_date = False
else:
    if month in (1,3,5,7,8,10,12):
        max_date = 31
    elif month in (4,6,9,11):
        max_date = 30
    else:
        if (year % 4==0):
            max_date = 29
        else:
            max_date = 28

    if (date > max_date or date < 1):
        valid_date = False

if valid_date:
    if date < max_date:
        date += 1
    else:
        date = 1
        if month == 12:
            month = 1
            year += 1
        else:
            month += 1

    print(f"{date:02d}-{month:02d}-{year}")
else:
    print("Invalid Date")
```

Execution :

51.06

Home

Logout

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Support

6.1.1. Incremented Date

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. Valid Month: 1 to 12

2. Valid Day: Depends on the month and year

- January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
- April (4), June (6), September (9), November (11): 1 to 30 days
- February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year

3. Valid Year: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:
<DD>-<MM>-<YYYY>
- If the date is invalid, print: "Invalid Date".

Sample Test Cases

+

Debugger

nextDate....

```
1 date = int(input())
2 month = int(input())
3 year = int(input())
4
5 valid_date = True
6
7 if (year <= 0 or month > 12 or month < 1):
8     valid_date = False
9 else:
10     if month in (1,3,5,7,8,10,12):
11         max_date = 31
12     elif month in (4,6,9,11):
13         max_date = 30
14     else:
15         if (year % 4==0):
16             max_date = 29
```

Average time0.013 s12.60 ms

Maximum time0.049 s49.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 149 ms

Test case 213 ms

Test case 310 ms

Test case 410 ms

Test case 56 ms

Terminal

Test cases